# Job Leon Feldbrugge

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## Referees

**Prof. Neil Turok**, University of Edinburgh, Edinburgh, United Kingdom and Perimeter Institute for Theoretical Physics, Waterloo, Canada (neil.turok@ed.ac.uk)
**Prof. Ue-Li Pen**, Academia Sinica, Taipei, Taiwan and Canadian Institute for Theoretical Astrophysics, Toronto, Canada (pen@cita.utoronto.ca)
**Prof. Celine Boehm**, University of Sydney, Sydney, Australia (celine.boehm@sydney.edu.au)
**Prof. Rien van de Weygaert**, University of Groningen, Groningen, The Netherlands (weygaert@astro.rug.nl)

## Academic positions

2026 - present **Member and Postdoctoral Fellow** (joint position) at the Institute for Advanced Study, Princeton, United States of America and the Leung Center for Cosmology and Particle Astrophysics, National Taiwan University, Taipei, Taiwan

2021 - 2026 **Higgs Fellow** at the Higgs Centre for Theoretical Physics at the University of Edinburgh, Edinburgh, United Kingdom

2019 - 2021 **Postdoctoral Fellow** (joint position) at the Perimeter Institute, Waterloo, Canada and Carnegie Mellon University, Pittsburgh, United States of America

## Areas of specialization

Applied mathematics: asymptotic analysis and resurgence, catastrophe theory, oscillatory integrals and Picard–Lefschetz theory • Scientific computing and machine learning • Wave optics and inverse problems • Lagrangian fluid dynamics and the cosmic web • Real-time path integrals and mathematical physics

## Education

2015 - 2019 **PhD** Physics, Perimeter Institute, University of Waterloo
*Advisor*: Neil Turok
*Thesis*: Path integrals in the sky: Classical and quantum problems with minimal assumptions
*Defense date*: October 17, 2019

2014 - 2015 **Master** Part III Mathematics (*with distinction*), University of Cambridge
*Essay supervisors*: Paul Shellard and Tommaso Giannantonio
*Essay*: Primordial non-Gaussianity and large-scale structure

2012 - 2014 **Master** Physics (*cum laude*), Van Swinderen Institute, University of Groningen

2012 - 2014 **Master** Astronomy (*cum laude*), Kapteyn Institute, University of Groningen

2012 - 2014 **Master** Mathematics (*cum laude*), Bernoulli Institute, University of Groningen
*Committee*: Rien van de Weygaert (cosmology and large-scale structure formation)
Diederik Roest (string cosmology)
Aernout van Enter (statistical mechanics)
*Thesis*: Statistics of caustics in large-scale structure formation

2009 - 2012 **Bachelor** Physics (*cum laude*), Van Swinderen Institute, University of Groningen

2009 - 2012 **Bachelor** Astronomy (*cum laude*), Kapteyn Institute, University of Groningen

2009 - 2012 **Bachelor** Mathematics (*cum laude*), Bernoulli Institute, University of Groningen

*Committee*: Rien van de Weygaert (cosmology and large-scale structure formation)
Elisabetta Pallante (quantum field theory)
Gert Vegter (computational geometry)
*Thesis*: Analysis of Betti numbers and persistence diagrams of 2D Gaussian random fields

July 2016 **Summer school** It from qubit summer school, Perimeter Institute, Waterloo, Canada (two weeks)
July 2015 **Summer school** Prospects in theoretical physics: new insights into quantum matter, Institute for Advanced Study, Princeton (one week)
July 2015 **Summer school** Princeton summer school on condensed matter physics, Princeton University and Institute for Advanced Study, Princeton (one week)
August 2011 **Summer school** Dealing with environmental heritage, Bath, United Kingdom (two weeks)
July 2010 **Summer school** University of Cambridge international summer school in science (two weeks)

## Awards

April 2020 Canadian Association of Physicists' Division of Theoretical Physics (DTP) and Winnipeg Institute for Theoretical Physics (WITP) P.R. Wallace PhD Thesis Prize
Best PhD thesis in Theoretical Physics from a Canadian university (2020).
November 2014 De Zeeuw-Van Dishoeck Award 2014
Best master thesis in Astronomy from a Dutch university (2014).
July 2014 GUF-100 prize 2014
Best student in the Faculty of Mathematics and Natural Sciences at the University of Groningen (2014).
November 2011 Silver medal in the University Physics Competition 2011
A university competition in which groups of three physics students all over the world solve a problem and write an article in 48 hours.
September 2010 Young Talent encouragement prize 2010 in Physics
Prize awarded by the Koninklijke Hollandse Maatschappij der Wetenschappen (Royal Holland Society of Sciences and Humanities) for the best freshman Physics student at the University of Groningen in 2009-2010.

## Scholarships

March 2015 University of Waterloo scholarship, for PhD at the Perimeter Institute
July 2014 Hendrik Muller Fund 2014: Scholarship for excelling Dutch students
June 2014 VSB Fund 2014: Scholarship for Dutch students studying abroad

## Research visits

November 2025 Research visit to Chris Howls and Inês Aniceto, University of Southampton, Southampton, United Kingdom
October 2025 Research visit to Sandrine Codis, Université Paris-Saclay, Paris, France
September 2025 Research visit to Chris Howls and Inês Aniceto, University of Southampton, Southampton, United Kingdom
May 2023 Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
March 2023 Research visit to Ue-Li Pen, Academia Sinica Institute of Astronomy and Astrophysics, Taipei, Taiwan
March 2023 Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
January 2023 Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
June 2022 Research visit to Rien van de Weygaert, Kapteyn Institute, Groningen, The Netherlands
January 2020 Research visit to Beatrice Bonga, Radboud University, Nijmegen, The Netherlands
June 2019 Research visit to Rien van de Weygaert, Kapteyn Institute, Groningen, The Netherlands
October 2018 Research visit to Rien van de Weygaert, Kapteyn Institute, Groningen, The Netherlands
May 2018 Research visit to Jean-Luc Lehners, Albert Einstein Institute, Potsdam, Germany
March 2017 Research visit to Jean-Luc Lehners, Albert Einstein Institute, Potsdam, Germany
Feb - Apr 2017 Research visit to DAMTP, University of Cambridge, United Kingdom

# Publications & talks

## Refereed journal articles and proceedings

2026 B. Hertzsch, **J. Feldbrugge**, and R. van de Weygaert, "Galaxy populations in the IllustrisTNG caustic skeleton," *Monthly Notices of the Royal Astronomical Society* 552, stag1724 (2026). arXiv:2604.18209 [astro-ph.CO]

2026 **J. Feldbrugge**, S. Crew, and U.-L. Pen, "Resurgence and hyperasymptotics in wave optics astronomy," *Journal of Cosmology and Astroparticle Physics* 2026(09), 022 (2026). arXiv:2602.21493 [astro-ph.CO]

2026 **J. Feldbrugge** and U.-L. Pen, "The real-time Feynman path integral for step potentials," *Annals of Physics* 490, 170490 (2026). arXiv:2508.17578 [quant-ph]

2026 A. Weber, **J. Feldbrugge**, and E. Pisanty, "Universal approach to saddle-point methods in attosecond science," *Physical Review A* 113, 063111 (2026). arXiv:2510.12545 [quant-ph]

2026 B. Hertzsch, **J. Feldbrugge**, M. Rodriguez, and R. van de Weygaert, "A new recipe for caustic pancakes: on the reality of walls in the cosmic web," *Journal of Cosmology and Astroparticle Physics* 2026(02), 037 (2026). arXiv:2510.02419 [astro-ph.CO]

2025 **J. Feldbrugge** and R. van de Weygaert, "What makes a cosmic filament? The dynamical origin and identity of filaments – I. Fundamentals in 2D," *Monthly Notices of the Royal Astronomical Society* 539, 873–901 (2025). arXiv:2405.20475 [astro-ph.CO]

2025 **J. Feldbrugge**, D. L. Jow, and U.-L. Pen, "Complex classical paths in quantum reflections and tunneling," *Physical Review D* 111, 085027 (2025). arXiv:2309.12420 [quant-ph]

2025 **J. Feldbrugge** and J. Y. L. Jones, "Efficient evaluation of real-time path integrals," *Physical Review D* 111, 083524 (2025). arXiv:2501.16323 [quant-ph]

2025 B. Bonga, **J. Feldbrugge**, and A. Metidieri, "Wave optics for rotating stars," *Physical Review D* 111, 063061 (2025). arXiv:2410.03828 [gr-qc]

2025 **J. Feldbrugge**, "Phase-space Delaunay tessellation field estimator," *Monthly Notices of the Royal Astronomical Society* 536, 807–815 (2025). arXiv:2402.16234 [astro-ph.CO]

2023 **J. Feldbrugge**, Y. Yan, and R. van de Weygaert, "Statistics of tidal and deformation eigenvalue fields in the primordial Gaussian matter distribution: the two-dimensional case," *Monthly Notices of the Royal Astronomical Society* 526, 5031–5053 (2023). arXiv:2301.07200 [astro-ph.CO]

2023 **J. Feldbrugge**, "Complex evaluation of angular power spectra: Going beyond the Limber approximation," *Physical Review D* 108, 103007 (2023). arXiv:2304.13064 [astro-ph.CO]

2023 D. L. Jow, U.-L. Pen, and **J. Feldbrugge**, "Regimes in astrophysical lensing: refractive optics, diffractive optics, and the Fresnel scale," *Monthly Notices of the Royal Astronomical Society* 525, 2107–2124 (2023). arXiv:2204.12004 [astro-ph.HE]

2023 **J. Feldbrugge** and N. Turok, "Existence of real time quantum path integrals," *Annals of Physics* 454, 169315 (2023). arXiv:2207.12798 [hep-th]

2023 **J. Feldbrugge**, "Multiplane lensing in wave optics," *Monthly Notices of the Royal Astronomical Society* 520, 2995–3006 (2023). arXiv:2010.03089 [astro-ph.CO]

2023 **J. Feldbrugge**, U.-L. Pen, and N. Turok, "Oscillatory path integrals for radio astronomy," *Annals of Physics* 451, 169255 (2023). arXiv:1909.04632 [astro-ph.HE]

2023 **J. Feldbrugge** and R. van de Weygaert, "Cosmic web & caustic skeleton: non-linear constrained realizations – 2D case studies," *Journal of Cosmology and Astroparticle Physics* 2023(02), 058 (2023). arXiv:2212.07840 [astro-ph.CO]

2021 G. Wilding, K. Nevenzeel, R. van de Weygaert, G. Vegter, P. Pranav, B. J. T. Jones, K. Efstathiou, and **J. Feldbrugge**, "Persistent homology of the cosmic web – I. Hierarchical topology in ΛCDM cosmologies," *Monthly Notices of the Royal Astronomical Society* 507, 2968–2990 (2021). arXiv:2011.12851 [astro-ph.CO]

2019 A. Di Tucci, **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "Quantum incompleteness of inflation," *Physical Review D* 100, 063517 (2019). arXiv:1906.09007 [hep-th]

2019 **J. Feldbrugge**, M. van Engelen, R. van de Weygaert, P. Pranav, and G. Vegter, "Stochastic homology of Gaussian vs. non-Gaussian random fields: graphs towards Betti numbers and persistence diagrams," *Journal of Cosmology and Astroparticle Physics* 2019(09), 052 (2019). arXiv:1908.01619 [astro-ph.CO]

2019 P. Pranav, R. van de Weygaert, G. Vegter, B. J. T. Jones, R. J. Adler, **J. Feldbrugge**, C. Park, T. Buchert, and M. Kerber, "Topology and geometry of Gaussian random fields I: on Betti numbers, Euler characteristic, and Minkowski functionals," *Monthly Notices of the Royal Astronomical Society* 485, 4167–4208 (2019). arXiv:1812.07310 [astro-ph.CO]

2018 **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "Inconsistencies of the new no-boundary proposal," *Universe* 4, 100 (2018). arXiv:1805.01609 [hep-th]

2018 **J. Feldbrugge**, R. van de Weygaert, J. Hidding, and Joost Feldbrugge, "Caustic skeleton & cosmic web," *Journal of Cosmology and Astroparticle Physics* 2018(05), 027 (2018). arXiv:1703.09598 [astro-ph.CO]

2018 **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "No rescue for the no boundary proposal: Pointers to the future of quantum cosmology," *Physical Review D* 97, 023509 (2018). arXiv:1708.05104 [hep-th]

2017 **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "No smooth beginning for spacetime," *Physical Review Letters* 119, 171301 (2017). arXiv:1705.00192 [hep-th]

2017 **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "Lorentzian quantum cosmology," *Physical Review D* 95, 103508 (2017). arXiv:1703.02076 [hep-th]

2016 **J. Feldbrugge**, J. Hidding, and R. van de Weygaert, "Statistics of caustics in large-scale structure formation," *The Zeldovich Universe: Genesis and Growth of the Cosmic Web*, IAU Symposium 308, 107–114 (2016). arXiv:1412.5121 [astro-ph.CO]

2011 R. van de Weygaert, G. Vegter, H. Edelsbrunner, B. J. T. Jones, P. Pranav, C. Park, W. A. Hellwing, B. Eldering, N. Kruithof, E. G. P. Bos, J. Hidding, **J. Feldbrugge**, E. ten Have, M. van Engelen, M. Caroli, and M. Teillaud, "Alpha, Betti and the megaparsec universe: on the topology of the cosmic web," *Transactions on Computational Science XIV*, Lecture Notes in Computer Science 6970, 60–101 (2011). arXiv:1306.3640 [astro-ph.CO]

### Preprints

2026 S. Crew and **J. Feldbrugge**, "Asymptotic evaluation of multidimensional oscillatory integrals" (2026). arXiv:2610.22767 [math-ph]

2026 I. Aniceto, **J. Feldbrugge**, and C. J. Howls, "Which saddles contribute? The south-east rule for multidimensional integrals" (2026). arXiv:2606.28271 [math-ph]

2026 A. Read, **J. Feldbrugge**, C. Boehm, R. van de Weygaert, and B. Hertzsch, "Caustic skeleton and the local cosmic web: the Coma cluster node and the Pisces-Perseus ridge" (2026). arXiv:2604.22213 [astro-ph.CO]

2023 **J. Feldbrugge**, D. L. Jow, and U.-L. Pen, "Crossing singularities in the saddle point approximation" (2023). arXiv:2309.12427 [quant-ph]

2023 **J. Feldbrugge** and N. M. D. Niezink, "Orthogonality relations for conical functions of imaginary order" (2023). arXiv:2309.05616 [math.CA]

2020 **J. Feldbrugge** and N. Turok, "Gravitational lensing of binary systems in wave optics" (2020). arXiv:2008.01154 [gr-qc]

### Theses

2019 **J. Feldbrugge**, "Path integrals in the sky: Classical and quantum problems with minimal assumptions," PhD thesis, Perimeter Institute, University of Waterloo, supervised by N. Turok. Available online.

2015 **J. Feldbrugge**, "Primordial non-Gaussianity and large-scale structure," Part III Essay, University of Cambridge, supervised by P. Shellard and T. Giannantonio. Available online.

2014 **J. Feldbrugge**, "Statistics of caustics in large-scale structure formation," Master thesis, University of Groningen, supervised by R. van de Weygaert, D. Roest, A.E. van Enter. Available online.

2012 **J. Feldbrugge** and M. van Engelen, "Analysis of Betti numbers and persistence diagrams of two-dimensional Gaussian random fields," Bachelor thesis, University of Groningen, supervised by R. van de Weygaert, E. Pallante, G. Vegter. Available online.

### Invited talks

October 2026 Filaments in the multi-scale Universe: understanding their role and detectability, Kavli IPMU, Kashiwa, Japan
Presentation: Caustic walls and filaments in the cosmic web

August 2026 Astronomy Department of The Physics Institute of the Federal University of Rio Grande do Sul, Porto Alegre, Brazil
Seminar: Unravelling the cosmic web with caustics

July 2026 EXCOSM workshop: Random Structures and Patterns in Spatial Data: Here, There, and Everywhere, Nancy, France
Presentation: The caustic skeleton of the cosmic web

May 2026 School of Physics and Astronomy, University of Nottingham, Nottingham, United Kingdom
Seminar: Real-Time Path Integrals, Caustics and Interference in Cosmology

May 2026 The Center of Gravity – Niels Bohr Institute, Copenhagen, Denmark
Seminar: Real-Time Path Integrals, Caustics and Interference in Cosmology

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| April 2026 | Cosmological Frontiers in Fundamental Physics 2026, Perimeter Institute, Waterloo, Canada<br>Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology |
| December 2025 | King's College London, London, United Kingdom<br>Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology |
| November 2025 | University of Southampton, Southampton, United Kingdom<br>Seminar: Unravelling the cosmic web with caustics |
| November 2025 | University of Vienna, Vienna, Austria<br>Presentation: Understanding the cosmic web with caustics |
| October 2025 | Université Paris-Saclay, Paris, France<br>Presentation: Understanding the cosmic web with caustics |
| October 2025 | University of Bonn, Bonn, Germany<br>Presentation: Real-Time Path Integrals, Caustics and Interference |
| October 2025 | Université Paris-Saclay, Paris, France<br>Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology |
| September 2025 | Cosmology seminar at Perimeter Institute for Theoretical Physics, Waterloo, Canada<br>Presentation: Real-Time Path Integrals, Caustics and Interference in Cosmology |
| August 2025 | Seminar at the Canadian Institute for Theoretical Astrophysics, Toronto, Canada<br>Presentation: Understanding the cosmic web with caustics |
| June 2025 | The Cosmic Web from Galaxies to Cosmology, Institute for Fundamental Physics of the Universe, Trieste, Italy<br>Presentation: Caustics and Interference |
| March 2025 | Seminar at the University of Pennsylvania, Philadelphia, United States<br>Presentation: Real-time Path Integrals, Caustics and Interference |
| October 2024 | Complexity and Cosmos, Gran Sasso Science Institute, L'Aquila, Italy<br>Presentation: Path integrals in the sky |
| June 2024 | Singular and Oscillatory Integrals, University College London, London, United Kingdom<br>Presentation: Integration in the complex plane with Picard-Lefschetz theory |
| May 2024 | Tuorla-Tartu meeting 2024: Borderless Universe, Tuorla, Finland<br>Presentation: What makes a cusp/filament? |
| December 2023 | Gauge Gravity by the Ghats seminar, Center for High Energy Physics (CHEP), Indian Institute of Science, Bengaluru, India<br>Presentation: Complex classical paths in quantum reflections and tunnelling |
| December 2023 | Large-scale parity violation, Academia Sinica Institute of Astronomy and Astrophysics, Taipei, Taiwan<br>Presentation: Dissecting the cosmic web with caustics |
| October 2023 | International Loop Quantum Gravity Seminar (online)<br>Presentation: Complex saddle points in gravitational path integrals |
| September 2023 | Theoretical Physics seminar, Newcastle University, Newcastle, United Kingdom<br>Presentation: On the existence of real-time path integrals |
| September 2023 | Complexity and Cosmos, Gran Sasso Science Institute, L'Aquila, Italy<br>Presentation: On the existence of real-time path integrals |
| July 2023 | Quantum Gravity 2023, Radboud University, Nijmegen, The Netherlands<br>Presentation: On the existence of real-time path integrals |
| July 2023 | Theoretical physics group, University of New Brunswick, Fredericton, Canada<br>Presentation: On the existence of real-time path integrals |
| March 2023 | Optimal Transport Theory and Applications to Physics, École de Physique des Houches, France<br>Presentation: Dissecting the cosmic web with caustics |
| September 2022 | 2nd Roman Juszkiewicz Symposium, Nicolaus Copernicus Astronomical Center, Warsaw, Poland<br>Presentation: Dissecting the cosmic web with caustics |
| June 2022 | Information Universe 4, University of Groningen, Groningen, The Netherlands<br>Presentation: Dissecting the cosmic web with caustics |
| May 2022 | Cosmology seminar, University of Oxford, Oxford, United Kingdom<br>Presentation: Dissecting the cosmic web with caustics |
| May 2022 | UK Cosmo Meeting 2022, Newcastle University, Newcastle, United Kingdom<br>Presentation (keynote): Dissecting the cosmic web with caustics |
| October 2021 | Higgs Hour, University of Edinburgh, Edinburgh, United Kingdom<br>Presentation: Interference, caustics and oscillatory integrals |
| May 2021 | Seminar, Universidad Nacional Autónoma de México, Mexico City, Mexico<br>Presentation: Interference phenomena in lensing and quantum physics |
| May 2021 | Sirius A symposium 2021: To infinity and beyond, Groningen, The Netherlands<br>Presentation: The caustic skeleton of the cosmic web |

October 2020 Quantum & Gravity Seminar, Radboud University, Nijmegen, The Netherlands
Presentation: Lorentzian quantum cosmology

October 2020 Pulsar group meeting, CITA, Toronto, Canada
Presentation: Multi-plane lensing and gravitational binary lensing in wave optics

November 2019 Scintillometry 2019, Max Planck Institute for Radio Astronomy, Bonn, Germany
Presentation: Oscillatory path integrals for radio astronomy

September 2016 CITA-PI day: Gravitational non-linear instability, CITA, Toronto, Canada
Presentation: Shocks in the early universe and gravitational waves

### Contributed talks

June 2025 Cosmology from Home 2025 (online)
Presentation: What makes a cosmic wall/filament in the caustic skeleton of the cosmic web?

June 2025 Triangular Conference on Cosmological Frontiers in Fundamental Physics 2025, Paris, France
Poster: Real-Time Path Integrals

October 2024 Scintillometry 2024, University of Central Florida, Orlando, United States
Presentation: Lensing by rotating stars

July 2024 Relativistic effects and novel observables, University of Geneva, Geneva, Switzerland
Presentation: What makes a filament/wall?

July 2024 Cosmology from Home 2024 (online)
Presentation: What makes a filament/wall?

June 2024 Theoretical Modeling of the Large Scale Structure of the Universe, University of Edinburgh, Edinburgh, United Kingdom
Presentation: What makes a filament/wall?

March 2024 Large Scale Structure group meeting, University of Cambridge, Cambridge, United Kingdom
Presentation: Caustic skeleton: what makes a filament in the cosmic web?

February 2023 The Co-evolution of the Cosmic Web and Galaxies across Cosmic Time, Kavli Institute for Theoretical Physics, UC Santa Barbara, California, United States
Presentation: Dissecting the cosmic web with caustics

July 2022 Cosmology from Home 2022 (online)
Presentation: Dissecting the cosmic web with caustics

March 2022 Cosmic Cartography 2022: Exploring the Cosmic Web and Large-Scale Structure, Kavli IPMU, Kashiwa, Japan
Presentation: The caustic web and non-linear constrained Gaussian random fields

October 2021 Tuorla-Tartu meeting 2021, University of Turku, Turku, Finland
Presentation: Caustic skeleton of the cosmic web

September 2020 Cosmology from Home 2020 (online)
Presentation: The caustic skeleton of the cosmic web

March 2020 Topological statistics group meeting, Department of Statistics, Carnegie Mellon University, Pittsburgh, United States
Two presentations: Cosmology and topology I, and Cosmology and topology II

February 2020 Centre for the Universe and Waterloo Centre for Astrophysics day, Waterloo, Canada
Presentation: Path integrals for radio astronomy and gravitational lensing II

February 2020 Cosmology group meeting, Perimeter Institute, Waterloo, Canada
Presentation: The caustic skeleton of the cosmic web

January 2020 The cosmic web in the local universe, Lorentz Center, Leiden, The Netherlands
Presentation: The caustic skeleton of the cosmic web

October 2019 The Future of Astronomy, Waterloo Centre for Astrophysics, Waterloo, Canada
Poster: Oscillatory path integrals for radio astronomy

September 2019 Theory group meeting, Carnegie Mellon University, Pittsburgh, United States
Presentation: Interference and Picard-Lefschetz theory

September 2019 Simplicity III, Perimeter Institute, Waterloo, Canada
Presentation: Fun with path integrals II

August 2019 Graduate student meeting, Perimeter Institute, Waterloo, Canada
Presentation: Oscillatory integrals in the complex plane

June 2019 Probabilities in cosmology, University of Groningen, The Netherlands
Presentation: Lorentzian beginnings of the universe

June 2019 The cosmic web: from galaxies to cosmology, Edinburgh, United Kingdom
Presentation: Caustic skeleton of the cosmic web

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| May 2019 | Cosmology group meeting Perimeter Institute, Waterloo, Canada<br>Presentation: Lenses and oscillatory integrals |
| May 2018 | Albert Einstein Institute group meeting, Potsdam, Germany<br>Presentation: Classical and weak trajectories |
| November 2017 | Path integral of gravity, Perimeter Institute, Waterloo, Canada<br>Presentation: Quantum incompleteness of inflation II |
| September 2017 | Cosmology group meeting CITA, Toronto, Canada<br>Presentation: The instability of the no-boundary proposal |
| July 2017 | Cosmic web day, University of Toronto, Toronto, Canada<br>Presentation: The caustic skeleton of the cosmic web |
| May 2017 | PI-day, Perimeter Institute, Waterloo, Canada<br>Presentation: Lorentzian quantum cosmology |
| May 2017 | Theory Canada 12, York University, Toronto, Canada<br>Presentation: Lorentzian quantum cosmology |
| May 2017 | Cosmology group meeting Perimeter Institute, Waterloo, Canada<br>Presentation: Caustics in large-scale structure |
| May 2017 | String cosmology group meeting Van Swinderen Institute, Groningen, The Netherlands<br>Presentation: Lorentzian quantum cosmology |
| April 2017 | New Thoughts 3: About the universe and more, Ely, United Kingdom<br>Presentation: Lorentzian quantum cosmology |
| April 2017 | British gravity meeting 2017, University of Oxford, Oxford, United Kingdom<br>Presentation: Lorentzian quantum cosmology |
| June 2016 | Cosmology group meeting CITA, Toronto, Canada<br>Presentation: Statistics of caustics in large-scale structure |
| May 2016 | Cosmology group meeting Perimeter Institute, Waterloo, Canada<br>Presentation: Statistics of caustics in large-scale structure |
| March 2016 | Statistics of extrema of large-scale structure, Lorentz Center, Leiden, The Netherlands<br>Presentation: Statistics of caustics in large-scale structure |
| June 2014 | IAUS 308: The Zel'dovich universe, Tallinn, Estonia<br>Presentation: Statistics of caustics in large-scale structure formation |
| October 2012 | Structure of the cosmic web, Leibniz Institute for Astrophysics, Potsdam, Germany<br>Presentation: Analysis of Betti numbers and persistence diagrams in 2D GRFs |

### Conferences and workshops (attended only)

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| March 2026 | Max Planck-IAS-NTU Center Kick-off Conference, Institute for Advanced Study, Princeton, United States |
| August 2025 | Charting the Future Symposium: Big questions in particle physics, strong gravity, and cosmology over the next 25 years, Perimeter Institute, Waterloo, Canada |
| June 2022 | Analogue Models of Gravity and Fluctuation-Induced phenomena, University of Edinburgh, Edinburgh, United Kingdom |
| June 2022 | Online workshop: Physics of the Early Universe |
| January 2021 | Cosmology 2021: the rise of field theory, University of Cambridge, Cambridge, United Kingdom |
| October 2020 | The information universe: What is the role of information in our Universe? University of Groningen, Groningen, The Netherlands |
| January 2020 | First Dutch Mathematical Relativity Day, Radboud University Nijmegen, Nijmegen, The Netherlands |
| September 2019 | Cosmological frontiers in fundamental physics 2019, Perimeter Institute, Waterloo, Canada |
| November 2018 | Quantum universe, in celebration of Neil Turok's 60th birthday, Centro de Estudios Científicos (CECs), Valdivia, Chile |
| August 2018 | Cosmology and gravitational physics with lambda, Nordita, Stockholm, Sweden |
| June 2018 | Scanning new horizons: Emergent space-time, black holes and quantum information, Van Swinderen Institute, Groningen, The Netherlands |
| January 2018 | Gravity in the early universe, Princeton University, Princeton, United States |
| June 2017 | Bounce scenarios in cosmology, Perimeter Institute, Waterloo, Canada |
| January 2017 | Fundamentals of the universe, Van Swinderen Institute, Groningen, The Netherlands |
| January 2017 | New directions in theoretical physics II, Higgs Centre for Theoretical Physics, Edinburgh, United Kingdom |
| October 2016 | Midwest relativity meeting, Perimeter Institute, Waterloo, Canada |
| June 2016 | Time in cosmology, Perimeter Institute, Waterloo, Canada |
| June 2016 | Concepts and paradoxes in a quantum universe, Perimeter Institute, Waterloo, Canada |
| June 2016 | Cosmological frontiers in fundamental physics 2016, Perimeter Institute, Waterloo, Canada |
| June 2015 | Convergence, Perimeter Institute, Waterloo, Canada |

April 2014 Quantum universe 4, University of Groningen, The Netherlands
March 2013 Quantum universe 3, University of Groningen, The Netherlands

## Software

Open-source codes that accompany my research.
**IntegralResurgence.jl** — Asymptotic evaluation of multidimensional oscillatory integrals using resurgence: Stokes constants, the adjacency of saddle points, and superasymptotic and hyperasymptotic approximations. Accompanies arXiv:2610.22767. Registered in the Julia General registry. GitHub · Documentation · Zenodo
**PicardLefschetzIntegration.jl** — Numerical evaluation of multidimensional oscillatory integrals using Picard–Lefschetz theory, with applications to wave optics in astronomy. Registered in the Julia General registry. GitHub · Documentation · Zenodo
**Picard–Lefschetz Path Integrals** (C++) — Evaluation of multidimensional oscillatory integrals using Picard–Lefschetz theory, demonstrated with lensing problems in wave optics; with U.-L. Pen and N. Turok. Accompanies Annals of Physics 451, 169255 (2023). Website
**PhaseSpaceDTFE.jl** — Density and velocity fields of $N$-body simulations with the phase-space Delaunay tessellation field estimator. Accompanies MNRAS 536, 807 (2025). Registered in the Julia General registry. GitHub · Documentation · Zenodo
**Nbody2D.jl** — Particle-mesh code for two-dimensional cosmological $N$-body simulations with unconstrained and constrained Gaussian random field initial conditions. Registered in the Julia General registry. GitHub · Documentation · Zenodo
**DiscreteNaturalNeighbors.jl** — Discrete natural neighbour interpolation of $n$-dimensional data, developed by Benjamin Hertzsch in collaboration with me. Registered in the Julia General registry. GitHub · Documentation
**HHGsimPL.jl** — Simulation of high-order harmonic generation using Picard–Lefschetz integration methods, developed by Anne Weber in collaboration with me. Accompanies Physical Review A 113, 063111 (2026). GitHub · Zenodo

## Advising

### PhD projects

2025 - present Advisor PhD project, Maé Rodriguez, University of Edinburgh, Edinburgh, United Kingdom: **Evaluating real-time path integrals for relativistic systems**
2025 - present Advisor PhD project, Amelie Read, University of Sydney, Sydney, Australia: **The caustic skeleton of the local cosmic web**
2024 - 2025 Co-advisor PhD project, Leigh Preimesberger, Carnegie Mellon University, Pittsburgh, United States: **Machine learning inference of the cosmic web**
2023 - present Advisor PhD project, Benjamin Hertzsch, University of Edinburgh, Edinburgh, United Kingdom: **Walls, filaments and galaxy populations in the caustic skeleton**
2023 - 2026 Co-advisor PhD project, Anne Weber, University College London, London, United Kingdom: **Complex trajectories in laser ionization and recombination experiments**
2023 - 2025 Co-advisor PhD project, Johanna Borissova, Perimeter Institute, Waterloo, Canada: **Lorentzian wormholes**
2023 - 2025 Co-advisor PhD project, Ariadna Metidieri, Radboud University, Nijmegen, The Netherlands: **Lensing of rotating stars and photons in wave optics**
2022 - present Co-advisor PhD project, Joshua Jones, Dublin Institute for Advanced Studies, Dublin, Ireland: **Real-time path integrals**
2020 - 2022 Co-advisor PhD project, Georg Wilding, University of Groningen, The Netherlands: **Topology of the two-dimensional cosmic web**
2020 - 2021 Co-advisor PhD project, Varun Rustagi, University of Groningen, The Netherlands: **The caustic skeleton of the local universe**

### Master projects

2026 - present Collaboration with Joanna Chan, independent scientist, Singapore: **Properties of the caustic skeleton in scale space**
2026 - present Collaboration with Guilherme Garcia, master student at the University of Copenhagen, Copenhagen, Denmark

2025 - 2026 Advisor master project, Guilherme Garcia, University of Edinburgh, Edinburgh, United Kingdom: **Constrained non-Gaussian fields**

2025 - 2026 Advisor master project, Joanna Chan, University of Edinburgh, Edinburgh, United Kingdom

2024 Advisor master project, Maé Rodriguez, University of Edinburgh, Edinburgh, United Kingdom: **Hamiltonian Monte Carlo sampling of primordial caustic constraints**

2023 Advisor master project, Vera Li, University of Edinburgh, Edinburgh, United Kingdom: **The caustic skeleton and redshift space distortions**

2022 - 2023 Advisor master project, Yihan Yan, University of Waterloo, Canada: **Statistics of critical points in eigenvalue fields**

2018 - 2022 Co-advisor master project, Kevin Bixerman, University of Groningen, The Netherlands: **The caustic skeleton of the local universe**

### Bachelor projects

2024 - 2025 Advisor bachelor project (Senior Honours), Guilherme Garcia, University of Edinburgh, Edinburgh, United Kingdom: **The cosmic web**

2022 - 2023 Co-advisor bachelor project, Yonatan Sklansky, University of Pennsylvania, United States: **Topology of multi-stream regions in $N$-body simulations**

2021 - 2022 Advisor bachelor project, Yihan Yan, University of Waterloo, Canada: **Homology of two- and three-dimensional Gaussian random fields**

## Teaching

July 2025 Lecturer, EXCOSM Summer School, Large-scale structure of the Universe: from galaxies to cosmology, Haapsalu, Estonia

June 2023 Lecturer, School on Fundamentals of the Universe, Lorentz Center, Leiden, The Netherlands

June 2018 2018 Rotman summer institute in philosophy of cosmology, Goderich, Canada
Two lectures on Lorentzian quantum cosmology

January 2018 PSI winter school, co-supervisor project 'Pair creation in de Sitter spacetime', Huntsville, Canada

2010 - 2015 Physics lecturer, exam training for secondary school students, UOCG Market BV

2013 - 2014 Developer of teaching material for exam training in physics, UOCG Market BV

2008 - 2014 Tutoring in mathematics and physics for secondary school students

## Service

2022 - 2023 Co-organizer of the conference New Directions in Theoretical Physics 4 (January 2023), University of Edinburgh, Edinburgh, United Kingdom

2018 - present Reviewer for Physical Review Letters (PRL), Physical Review D (PRD), the Journal of Cosmology and Astroparticle Physics (JCAP), the Journal of High Energy Physics (JHEP), Monthly Notices of the Royal Astronomical Society (MNRAS), and Universe.

2013 - 2014 Chairman of the Professor Hendrik de Waard Foundation

2012 - 2013 Treasurer of the Professor Hendrik de Waard Foundation

2011 - 2012 Member of the education committee mathematics, chairman of the student council

2010 - 2014 Guide at the Gratama telescope of the Blaauw observatory, Groningen, at stargazing events

## Languages

| | | | |
|---|---|---|---|
| Dutch | Mother tongue | English | Fluent |
| German | Elementary | French | Elementary |
| Latin | Elementary | | |

## Programming experience

| | | | |
|---|---|---|---|
| Mathematica | Fluent | C++ | Fluent |
| Python | Fluent | Matlab | Moderate |
| Swift | Moderate | Julia | Fluent |
| Fortran | Moderate | R | Elementary |